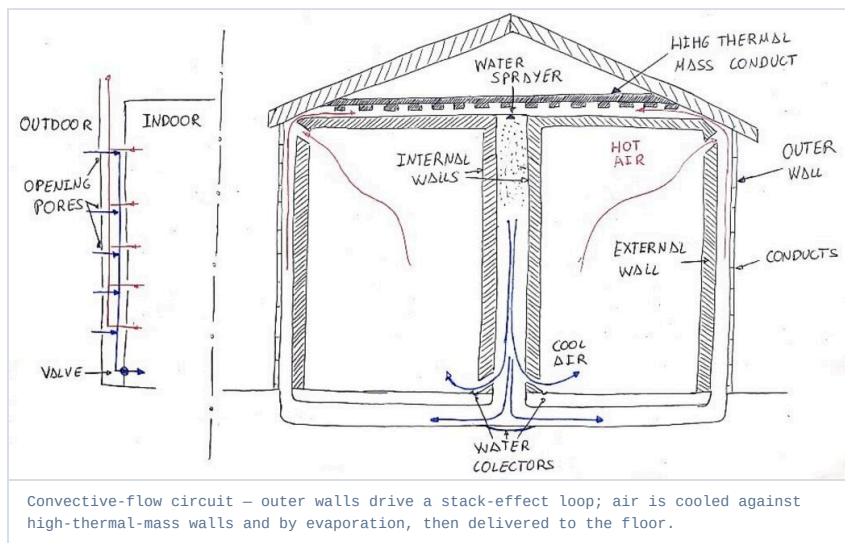


Passive Cooling from Termite Mounds

Reframing a big-building cooling principle into a self-regulating family house.

THE STUCK PROBLEM

Building cooling consumes a large share of global energy and emissions — and the existing termite-inspired designs exist only at large commercial scale. The challenge: extract the underlying principles and apply them to an ordinary family house that regulates its own temperature with little or no active cooling.



HOW THE THINKING WORKED

Extract. Distil the termite mound to its transferable mechanisms — heat-induced convection between low- and high-thermal-mass structures, and Venturi-driven induced flow — rather than copying its shape.

Translate. Re-express those as five concrete design features: an outer air-gap, high-thermal-mass internal walls and conduits, low-mass breathing outer walls, a connected chimney-and-flute network, and an optional water pulveriser for evaporative cooling.

CONCEPT & SANITY CHECK

The outer walls heat quickly and drive a stack-effect flow; air is cooled against the overnight-charged walls and by evaporation, then delivered to the floor. The loop reverses at night and stops in winter. Rather than asserting it would work, a first-order stack-effect calculation estimated a flow of about 0.46 m/s for a 100 m² house — enough to show viability and to flag a full thermo-fluids study as the honest next step.